



When comparing the values of efficiency displayed by the PPC6000 equipment and hand held equipment, **first** check that the following values are the same or very close:

- 1) Inlet temperature.
- 2) Flue gas temperature.
- 3) Calorific value
- 4) Hydro-carbon ratio.
- 5) Oxygen value (after making correction for dry to wet on the hand held).
- 6) Selection of net or gross.
- 7) If a radiated heat loss value is being included ensure the hand held equipment has corrected the value for the current firing rate of the burner.

Other points to remember:

In a typical package boiler radiation losses are usually in the order of 3%, blow down losses typically 2% and flue gas losses 15% to 20%.

Giving a typical overall efficiency of $100 - 3 - 2 - 20 = 75\%$.

The major area for attention to improve boiler efficiency is the flue gas losses, with the object of minimizing excess air, within the bounds of safe and reliable boiler operation. O2 Trim is the best tool for ensuring consistent optimum combustion and minimization of excess air.

The PPC6000 ensures consistent fuel: air ratios.

The Fireeye O2 Trim option continuously compensates for changes that affect efficient combustion due to such variables as: viscosity of liquid fuels, density of gaseous fuels, changes in temperature of fuels, changes in gas and oil pressures, worn and damaged burners, dirty burners, scaling of boiler tubes, ambient temperature, barometric pressure and tramp air.

9.6 Typical Wiring Diagrams

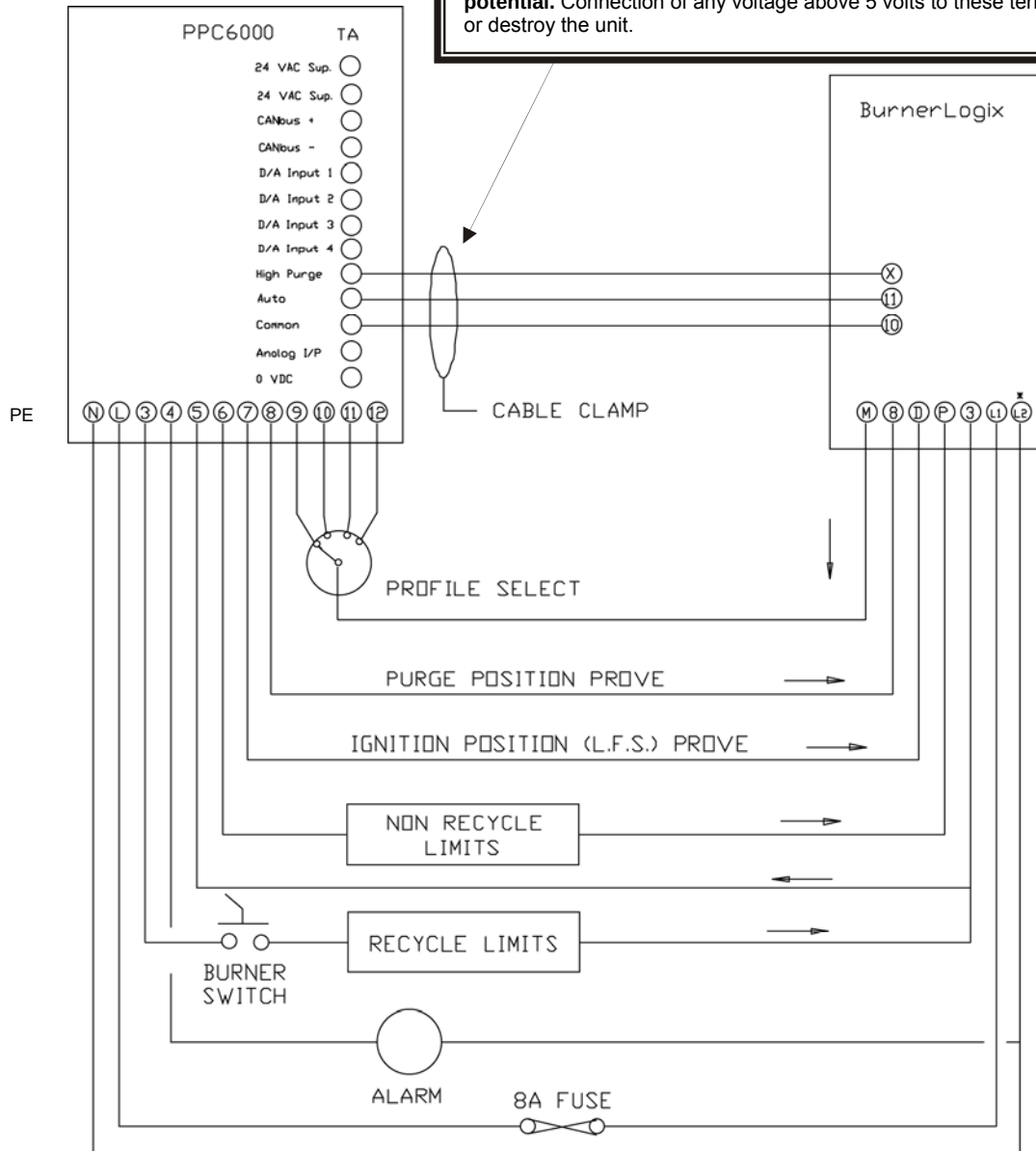
Typical Wiring Diagram / YB110

IMPORTANT: Low Voltage (vertical terminal strips)

All low voltage circuits and communication wire must be fully shielded braided type wire of the specified gauge and number of conductors. **Table 2.6.1-A provides the only approved wire for this application. No "or equal" is provided. Use of wire not approved by Fireye may VOID warranty.**

All wiring to terminal block "PA" & "PB" as well as to any optional daughter board (e.g. VSD) and the Power/CANbus wiring to the display, must be fully shielded braided wire per Table 2.6.1-A.

Under no circumstances should these input/outputs be connected to mains potential. Connection of any voltage above 5 volts to these terminals will damage or destroy the unit.



CAUTION

Due to the possibility of High Frequency electrical noise, **BOTH** ends of the shielded signal wire to the VSD **must** be grounded.

* Refer to bulletin BL-1001 for remaining BurnerLogix I/O connections. A complete schematic can be found at <http://fireve.net/pdf/JC1109.pdf>.

** **Note:** PE7 and PE8 provide purge and low fire position signals to the flame safeguard control. These outputs **MUST NOT** have a load greater than 30mA (i.e. relays, lamp, etc.), damage to the PPC6000 will result.



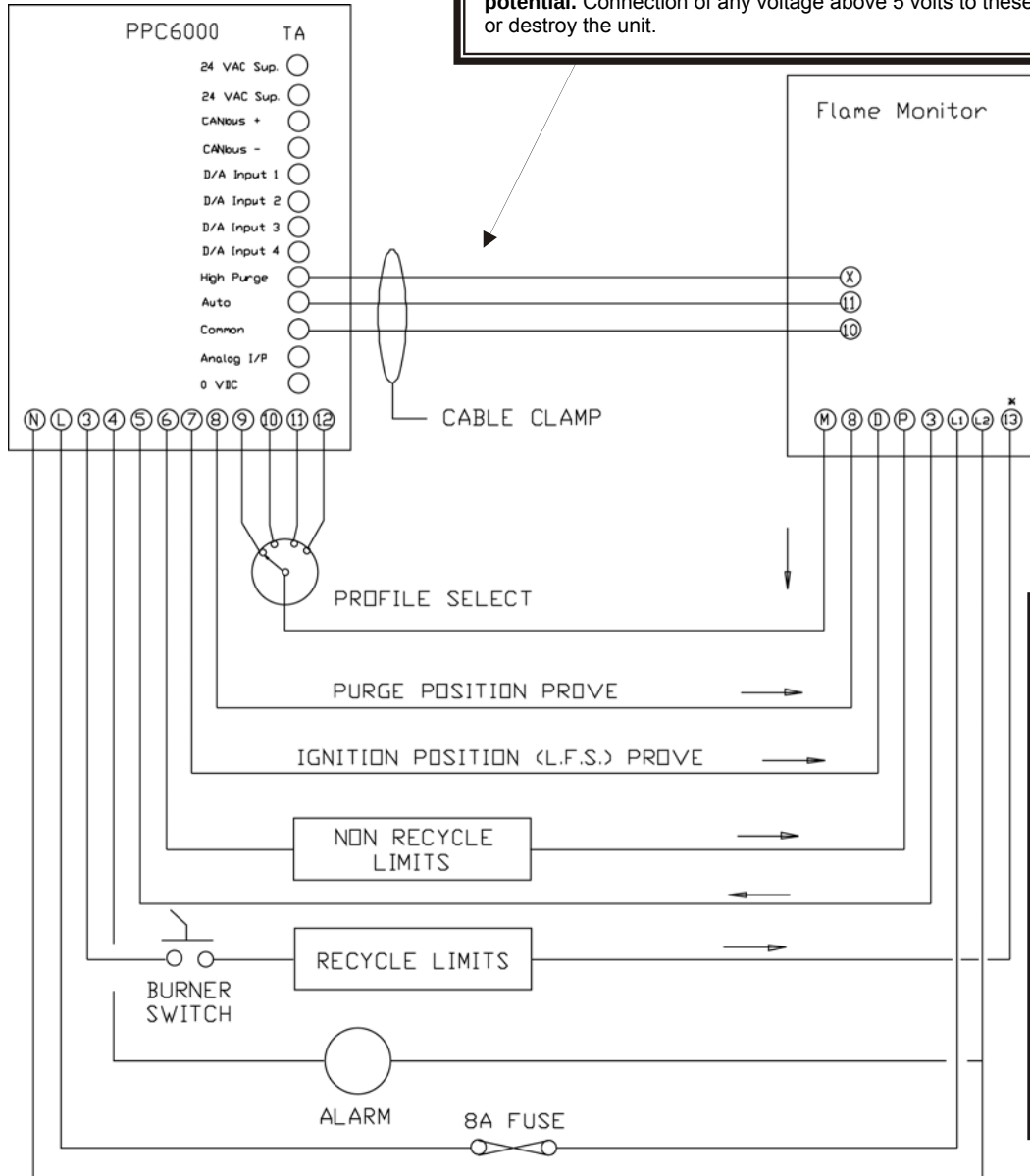
Typical Wiring Diagram / E110

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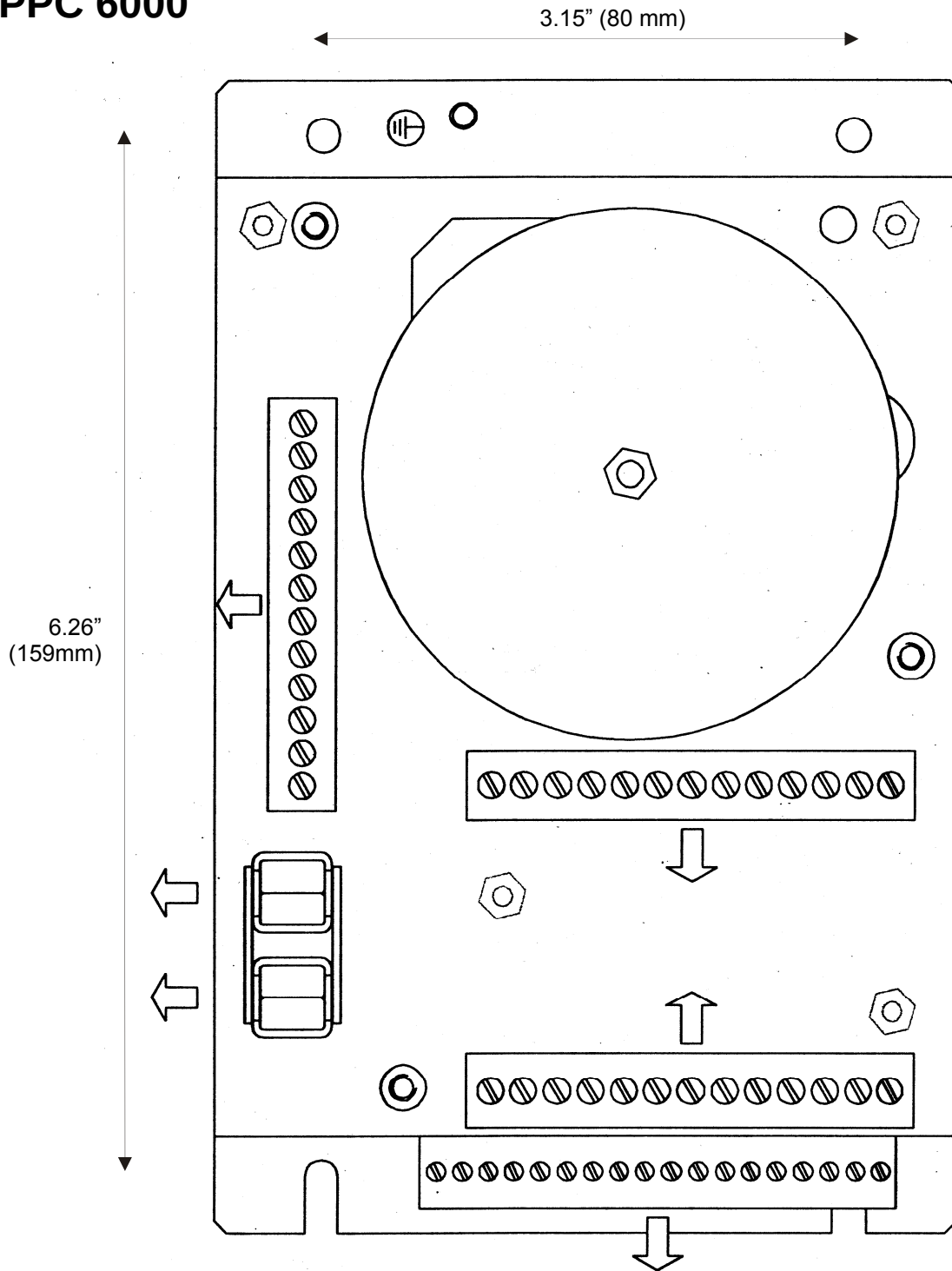
* Refer to bulletin E-1101 for remaining Flame Monitor I/O connections.
A complete schematic can be found at <http://fireeye.net/pdf/JC1109.pdf>.

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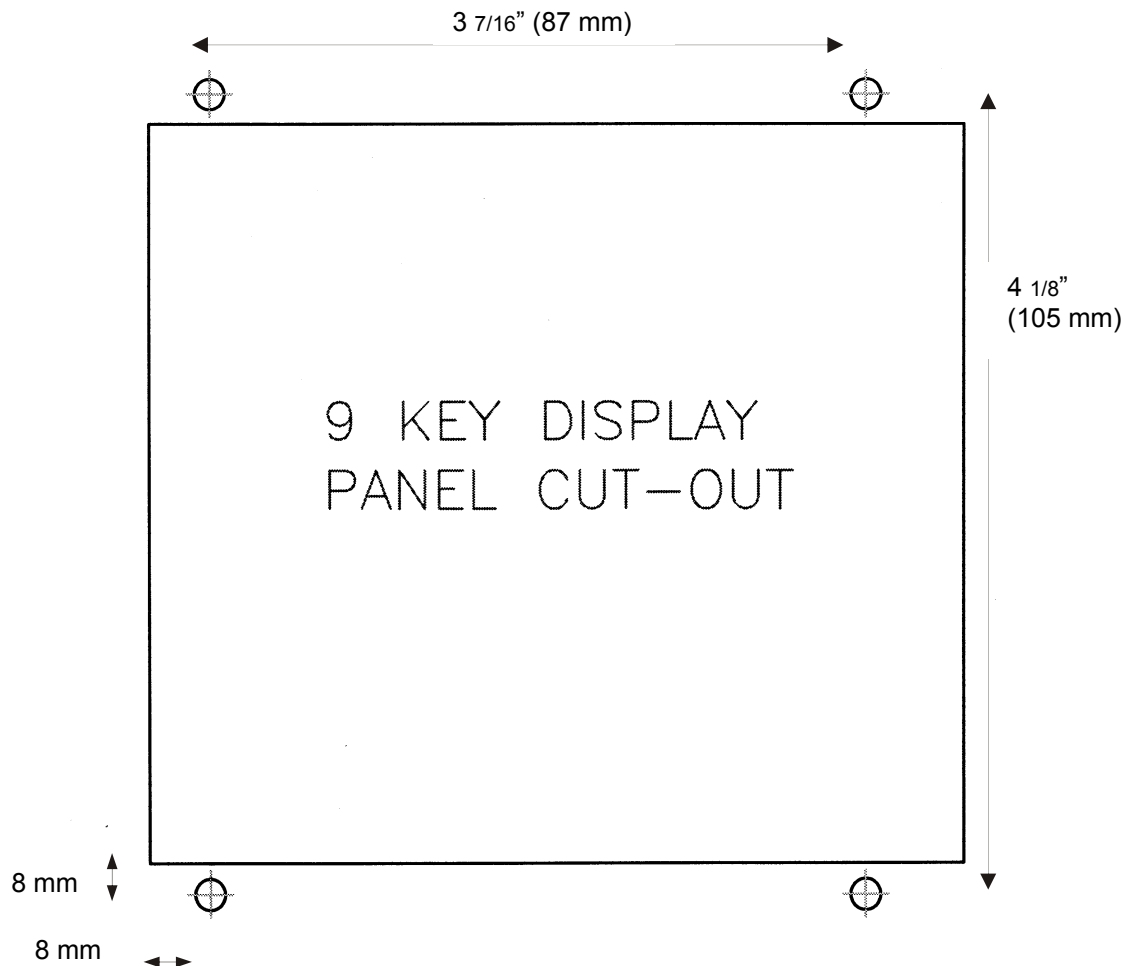
NOTE: These templates should be measured for accuracy as distortion from printing may occur. See section 2.0 for details.

PPC 6000





NX600, NX610

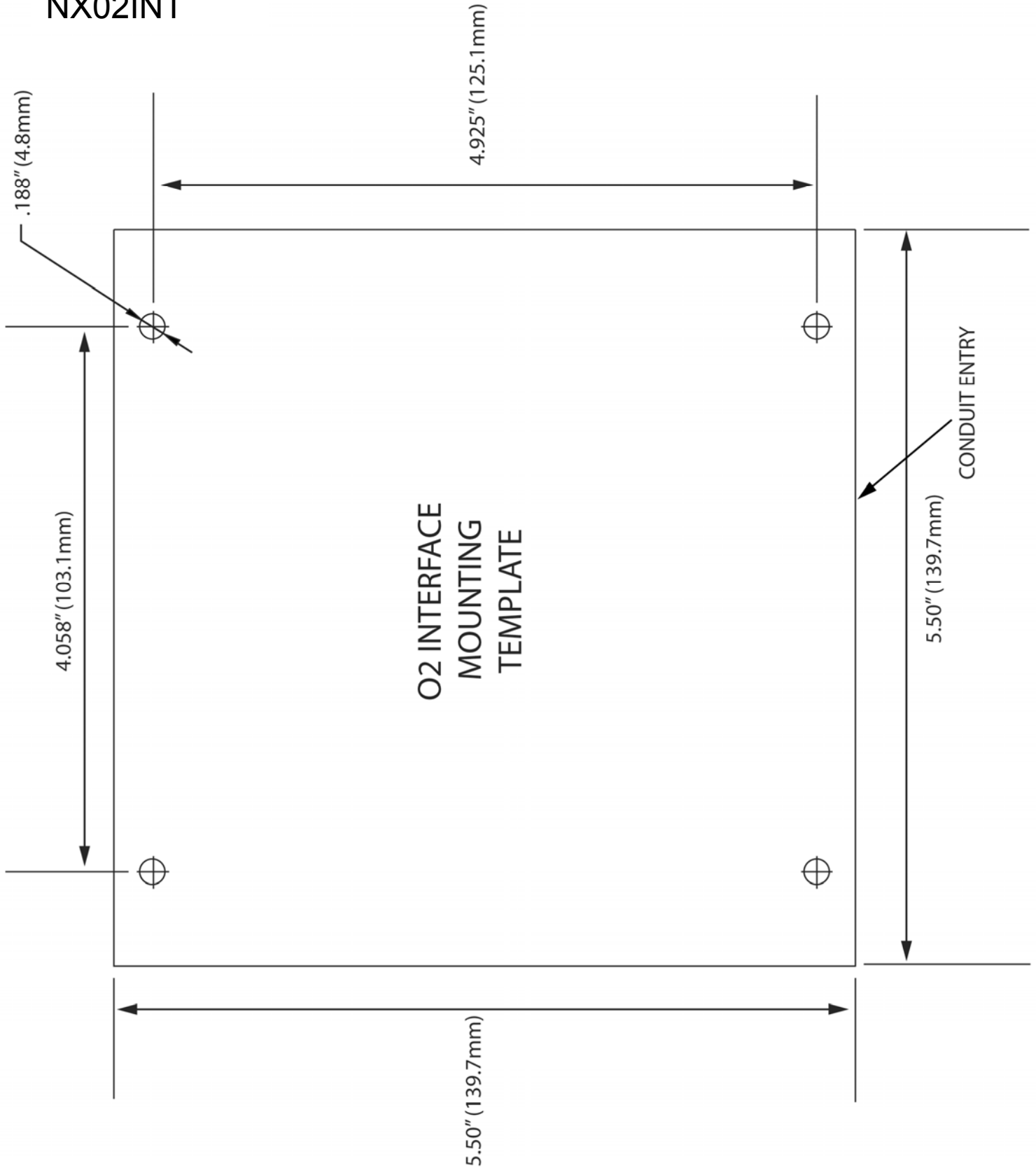


TEMPLATE: Remove to use

NOTE: These templates should be measured for accuracy as distortion from printing may occur. See section 2.0 for details.



NX02INT



TEMPLATE: Remove to use

NOTE: These templates should be measured for accuracy as distortion from printing may occur. See section 2.0 for details.



9.7 Operator's Guide

9.7.1 Introduction

This guide can be downloaded from the Fireeye web site at www.Fireeye.com. Search bulletins for NXOG-6001 under the Nexus products family.

As an operator there are generally only a few items of interest as far as operating the Nexus PPC6000. They are:

- Setting the system's operating pressure or temperature.
- Adjusting the cut in or cut out value.
- Manually modulating the boiler.
- Resetting the control after a safety shut down.
- Finding the Fault that caused the shut down.

All of the other options and adjustments should be done by the combustion technician, as changes to certain parameters could result in an unsafe operating mode.

9.7.2 General

The Nexus display and keypad allows the operator (or combustion technician) access to all setpoints and option parameters for commissioning and operating the Nexus PPC6000. The keys are labeled with their primary function on the key, their secondary function under the key. E.g.: COM/ENTER.

Several levels of passwords protect the integrity of the control. These passwords are numbers entered by using the up/down arrow keys after pressing the COM (commissioning) key. To enter the password start by pressing the COM (commissioning) key. You will have 30 seconds to reach the password value and press the ENTER (COM) key.

9.7.3 Password

The only password an operator should require is the "site Passcode." This can be set as a number from 0-999. The factory default number is 154. The site Passcode can be changed if desired, however, this requires the use of a higher-level Passcode.

Warning: Should the Passcode be changed and forgotten, the PPC6000 will have to be reprogrammed by using the higher-level Passcode. Until then, no adjustments are possible.

Operating Pressure – PID (setpoint) 1 or (2)*

*** NOTE: () Parentheses indicate setpoint 2 option parameter. E.g. Day/night.**

To set or adjust the boiler's operating pressure follow steps 1 thru 7:

1. Press the **COM** key.
2. Use the **UP** key to raise the value to **154** (or new Site Passcode, if changed).
3. Press the **ENTER** key. This is the same key as **COM**.
4. If the display shows the angular positions of the servomotors, press the **MODE** key to change to option parameters. Using the up or down key move to **OPTION 21.1 (22.1)**. This is the control value the Nexus will modulate the burner to maintain.



5. Use the left or right arrow key to move the cursor to the lower portion of the display, then using the up/down arrow key change the setting.
6. Press the **ENTER** key to accept the new setpoint.
7. To exit and return to the normal run mode, press the **RUN** key followed by the **ENTER** key.

Note: The next time you enter the site password you will return to the last option selected.

9.7.4 Adjusting Cut IN and Cut OUT

Follow steps 1 through 3 above.

4. If the display shows the angular positions of the servomotors, press the **MODE** key to change to option parameters. Using the up or down key, move to **OPTION 21.6 (22.6) for Low Limit (Cut In)** or **21.7 (22.7) for High Limit (Cut Out)**. This is the control value the Nexus will start/stop the burner after reaching the cut in/out value. This value is a differential value when option parameter 21.5 (22.5) is set at 2. That is, it is subtracted/added from/to the operating setpoint. Once set, this value will follow any change in the setpoint programmed in option 21.1 (22.1). These values are actual values if 21.5 (22.5) are set to 1. (see table below)

Option parameter 21.5 (22.5) value	Meaning
0	No limits. The burner will run until another method is used to switch it off.
1	Absolute limit. The values entered in option parameters 21.6 (22.6) and 21.7 (22.7) are the actual limit values.
2	Deviation limit. The values entered in option parameters 21.6 (22.6) and 21.7 (22.7) represent a deviation (i.e. offset) from the setpoint 1 control value. This means that if the setpoint control value is changed, the limits are automatically changed correspondingly.

5. Use the left or right arrow key to move the cursor to the lower portion of the display then - using the up/down arrow key, change the setting.
6. Press the **ENTER** key to accept the new setpoint.
7. To exit and return to the normal run mode press the **RUN** key followed by the **ENTER** key.

Deviation Example – setting Opt 21.5 to 2

To maintain 100 psi on the boiler, cut out at 115 psi, cut in at 95 psi.

NOTE: Opt 15.5 – Boiler High Safety Limit – If this option is set to a non-zero, *exceeding* this value causes a *non-volatile lockout*.

Opt 21.1 (22.1)* set to 100 psi
Opt 21.5 set to 2
Opt 21.6 (22.6)* set to 15
Opt 21.7 (22.7)* set to 5

RESULTS:

Boiler OFF (Cut Out) at.... 115 psi
Boiler ON (Cut In) at 95 psi
Boiler maintains (Setpoint)..... 100 psi via PID

Changing ONLY the Setpoint –

Opt 21.1 (22.2) change to 60 psi

NEW RESULTS from Setpoint Change: